

Proposal

Title

Climatological Characteristics of the Moisture Budget Associated with Mixed Rossby-Gravity Waves During the Indian Summer Monsoon

1. Scientific Question

How do the different moisture budget components evolve during the life cycle of Mixed Rossby-Gravity (MRG) waves over the Indian monsoon region, and what role do they play in initiating and maintaining deep convection?

2. Motivation

Convectively Coupled Equatorial Waves (CCEWs) play an important role in organizing tropical convection and modulating rainfall over the Indian Ocean and monsoon region. While Kelvin and Equatorial Rossby (ER) waves have been studied extensively, the thermodynamic characteristics of Mixed Rossby-Gravity (MRG) waves remain relatively unexplored, particularly over the Indian monsoon region.

Previous studies have shown that MRG waves influence convection through large-scale moisture convergence and dynamical forcing. However, the evolution of individual moisture budget components during different phases of MRG waves has not been systematically investigated.

Understanding these processes will improve our knowledge of tropical wave-convection interactions and provide a foundation for future evaluation of numerical weather prediction models.

3. Objectives

1. Identify climatological MRG wave activity during JJAS using filtered OLR.
2. Examine the spatial and temporal evolution of convection associated with MRG waves.
3. Compute the vertically integrated moisture budget associated with MRG waves.
4. Quantify the contribution of :
 - Horizontal moisture advection
 - Vertical moisture advection
 - Moisture convergence
 - Surface evaporation (if available)

during different phases of MRG waves.

5. Identify the dominant moisture processes responsible for the initiation and maintenance of convection.

4. Datasets

- NOAA CERES Daily OLR
- ERA5 Reanalysis
 - Specific humidity
 - Zonal and meridional wind
 - Vertical velocity
 - Surface latent heat flux (if required)
- IMERG precipitation (or IMD gridded rainfall)

5. Methodology

Step 1

Compute OLR anomalies and apply Wheeler-Kiladis filtering to isolate MRG waves.

Step 2

Identify MRG wave phases (active, peak and suppressed) over the Indian monsoon region during JJAS.

Step 3

Generate climatological composites of

- OLR
- Rainfall
- Low-level winds

for each wave phase.

Step 4

Compute vertically integrated moisture budget components from ERA5.

Step 5

Composite each moisture budget component with respect to the MRG wave life cycle.

Step 6

Investigate the temporal evolution between

- Moisture convergence
- Rainfall
- OLR

to determine whether moisture build up precedes deep convection.

6. Expected Outcomes

- Climatological structure of MRG waves over the Indian monsoon region.
- Evolution of moisture budget components during different phases of MRG waves.
- Identification of the dominant moisture process responsible for convection.
- Better understanding of moisture preconditioning prior to deep convection.
- A benchmark observational study that can later be extended for numerical model evaluation.

7. Tentative Figures

Figure 1

Wheeler-Kiladis spectrum showing MRG wave activity.

Figure 2

Composite evolution of

- OLR
- Rainfall
- Low-level circulation

through the MRG wave life cycle.

Figure 3

Composite maps of

- Moisture convergence
- Horizontal moisture advection
- Vertical moisture advection

for different wave phases.

Figure 4

Lag composites (or Hovmöller diagrams) showing

- Moisture convergence
- OLR
- Rainfall

relative to the MRG wave peak.

Figure 5

Schematic summarizing the evolution of moisture and convection during an MRG wave.